AquaFlow X200 Submersible Pump - User Manual

Model AQF-X200. Manual revision C, March 2025. Fictional product, written as sample data for DocuMind.

Important safety information

Read this manual completely before installing or using the pump, and keep it for future reference. Failure to follow these instructions can cause electric shock, fire, flooding or serious injury.

- **Disconnect the power** before handling, cleaning or servicing the pump. Never lift or lower the pump by its power cable; use the carrying handle or a rope attached to it.
- The pump must be connected through a residual current device (RCD) with a rated tripping current of no more than 30 mA.
- Do not use the pump to move flammable, explosive or corrosive liquids, fuel, oil, or drinking water.
- Do not use the pump in swimming pools or ponds while people or animals are in the water.
- Children must not play with the pump or carry out cleaning or maintenance.
- If the power cable is damaged it must be replaced by the manufacturer or an authorised service centre, never repaired with tape.

Intended use

The AquaFlow X200 is designed for draining flooded cellars, emptying water tanks and rain barrels, and transferring clean or slightly dirty water containing soft particles up to 25 mm in diameter. It is intended for domestic and light commercial use.

Technical specifications

Parameter	Value
Model number	AQF-X200
Rated power	1.1 kW (1,100 W)
Supply voltage	230 V AC, 50 Hz, single phase
Maximum flow rate	4,200 litres per hour
Maximum delivery head	38 m
Maximum immersion depth	7 m
Maximum particle size	25 mm
Maximum fluid temperature	35 degrees C
Protection rating	IPX8 (motor housing)
Insulation class	F
Outlet connection	1 inch BSP female, with 25/32/38 mm hose adapter
Power cable	10 m, type H07RN-F 3G1.0
Weight (without cable)	9.4 kg
Dimensions (H x D)	360 x 190 mm
Sound pressure level	below 55 dB(A) at 1 m

Performance curve summary

At a delivery head of 10 m the pump delivers approximately 3,100 litres per hour. At 20 m the output falls to about 2,000 litres per hour, and at 30 m to about 900 litres per hour. Flow stops at the maximum delivery head of 38 m. Long or narrow hoses reduce flow; for runs over 15 m use a 38 mm hose.

Package contents and installation

What is in the box

- AquaFlow X200 pump with integrated float switch and 10 m cable
- Multi-size hose adapter (25 / 32 / 38 mm) with sealing ring
- Non-return (check) valve
- 2 m lifting rope
- This manual and the warranty registration card

Positioning the pump

Place the pump upright on a firm, level surface. In a sump or tank with a muddy bottom, stand the pump on a brick or a flat stone so that it does not suck in sludge. The minimum water depth for starting the pump is 25 cm; in manual mode the pump can draw the water level down to 3 cm.

Make sure the float switch can move freely and does not touch the walls of the sump. The sump should be at least 40 x 40 cm.

Connecting the hose

1. Screw the non-return valve into the pump outlet using the supplied sealing ring.
2. Cut the hose adapter to match your hose diameter and screw it onto the valve.
3. Fix the hose with a hose clamp. Avoid kinks; a kinked hose causes overheating.

The non-return valve prevents water in the delivery pipe from flowing back when the pump stops, which reduces the number of start cycles and extends motor life.

Electrical connection

The AquaFlow X200 is supplied ready to plug in. Connect it only to a correctly installed, earthed socket protected by a residual current device of 30 mA maximum.

- The supply voltage must match the rating plate: 230 V AC, 50 Hz. A supply voltage below 190 V causes the controller to stop the pump with error code E05.
- The plug and socket must be installed in a flood-safe location, at least 1 m above the highest expected water level.
- Extension cables are not recommended. If unavoidable, use a cable of type H07RN-F with a cross-section of at least 1.5 square millimetres and a maximum length of 20 m, and keep the connection dry.
- Do not connect the pump to a generator smaller than 3 kVA. The starting current is approximately 3 times the rated current.

Fuse rating

Use a 10 A slow-blow (type C) circuit breaker. A fast-acting fuse may trip during start-up.

Earthing

The pump is protection class I and must always be earthed. Never remove or bypass the earth pin on the plug.

Operation

Control panel

The control panel on the pump head has a MODE button and a status LED.

LED	Meaning
Solid green	Ready, waiting for water (automatic mode)
Flashing green	Pump running
Solid blue	Manual mode active
Flashing red	Fault: see the error code table
Off	No power

Automatic mode

In automatic mode the integrated float switch starts the pump when the water level rises to 52 cm (switch-on level) and stops it when the level falls to 18 cm (switch-off level). This is the default mode and is recommended for flood protection in cellars.

Manual mode

Press MODE once to switch to manual mode. The pump then runs continuously and ignores the float switch, so you can drain down to 3 cm. In manual mode the dry-run protection still stops the pump if it runs without water for more than 90 seconds.

Dry-run protection

The controller monitors motor load. If it detects dry running, it stops the pump and shows error E01. In automatic mode it retries every 30 minutes, up to 5 times.

Duty cycle

The pump is rated for continuous operation when fully submerged. When the motor housing is not submerged, limit operation to 10 minutes per hour.

Maintenance and storage

Disconnect the power supply before any maintenance.

Regular maintenance schedule

Interval	Task
After each use with dirty water	Rinse the pump with clean water
Every 3 months	Remove and clean the intake filter (part AQ-FIL-90)
Every 6 months	Check the float switch moves freely; inspect the cable
Every 12 months	Inspect the impeller for wear; check the mechanical seal for leaks

Cleaning the intake filter

Turn the base plate a quarter turn anticlockwise and pull it off. Remove the filter cage and rinse it under running water. Do not use solvents. Refit the base plate until it clicks.

Impeller inspection

If flow has dropped noticeably, remove the base plate and check the impeller for fibres, stones or wear. A worn impeller should be replaced with part AQ-IMP-200.

Winter storage

If there is a risk of frost, remove the pump from the water, run it in manual mode in a bucket of clean water for one minute, then drain it completely. Store it upright in a dry place at a temperature above 5 degrees C. Ice inside the pump housing can crack the mechanical seal.

Troubleshooting

Symptom	Possible cause	Remedy
Pump does not start	No power, RCD tripped	Check the socket and RCD; reset the RCD
Pump does not start	Water level below switch-on level	Wait for the level to rise, or use manual mode
Pump runs but delivers no water	Air lock in the pump	Tilt the pump to let air escape; clean the vent hole
Pump runs but delivers no water	Hose kinked or check valve stuck	Straighten the hose; clean the non-return valve
Low flow rate	Intake filter blocked	Clean the intake filter
Low flow rate	Delivery head too high or hose too long	Use a wider hose or reduce the lift height
Pump starts and stops frequently	Sump too small, float switch catches	Use a larger sump; free the float switch
Pump stops after a short time	Thermal protection triggered	See error E03; let the motor cool
RCD trips when pump starts	Cable or motor insulation fault	Stop using the pump; contact service
Loud noise or vibration	Stone or debris in the impeller	Disconnect power and clean the impeller

If the problem persists after these checks, contact AquaFlow Service (details on the last page). Do not open the motor housing: doing so voids the warranty and may break the IPX8 seal.

Error codes

When the status LED flashes red, the controller shows an error code on the display. The pump stops until the fault is cleared.

Code	Meaning	What to do
E01	Dry run detected	Make sure the intake is submerged; clean the filter
E02	Overcurrent	Check for a blocked impeller or damaged cable; reset
E03	Motor overheating	The thermal cut-out has tripped at 80 degrees C winding temperature. Let the pump cool for 30 minutes
E04	Float sensor fault	Check the float switch moves freely; contact service if it persists
E05	Supply voltage too low (below 190 V)	Avoid long extension cables; check the supply
E06	Supply voltage too high (above 255 V)	Have the supply checked by an electrician
E07	Impeller blocked	Disconnect power and remove debris from the impeller
E09	Internal controller error	Contact service

Resetting an error

After fixing the cause, reset the controller by holding the MODE button for 5 seconds until the LED turns green. If the same error appears three times within one hour, the controller locks out and the pump must be unplugged for at least 60 seconds before it can be reset.

Error history

Hold MODE and press it twice quickly to scroll through the last 10 error codes. Service technicians use this history to diagnose recurring faults.

Spare parts and accessories

Use only genuine AquaFlow spare parts. Third-party parts may damage the pump and void the warranty.

Part number	Description
AQ-IMP-200	Vortex impeller, glass-fibre reinforced
AQ-SEAL-12	Mechanical shaft seal, silicon carbide
AQ-FLT-03	Float switch assembly with cable
AQ-FIL-90	Intake filter cage
AQ-CHK-25	Non-return valve, 1 inch
AQ-ADP-38	Hose adapter set (25/32/38 mm)
AQ-CAB-10	Replacement power cable, 10 m H07RN-F
AQ-CTL-X2	Controller board (service centre only)

Accessories

- **AQ-ALM-1 Water alarm:** a battery-powered sensor that beeps and sends a phone notification when water reaches the sensor.
- **AQ-HOSE-38 Flat hose:** 38 mm lay-flat hose, 15 m, with quick couplings.
- **AQ-BASE-5 Raised base:** lifts the pump 5 cm to avoid sucking in sludge.

Ordering

Parts can be ordered from authorised dealers or directly at parts.aquaflow.example. Have the serial number from the rating plate ready.

Warranty and service

Warranty period

The AquaFlow X200 comes with a 3-year manufacturer's warranty from the date of purchase. If you register the product at aquaflow.example/register within 60 days of purchase, the warranty is extended to 5 years.

What the warranty covers

The warranty covers defects in materials and workmanship. During the warranty period we will repair or replace the pump free of charge. Wear parts (impeller, filter cage and shaft seal) are covered for the first 12 months only.

What the warranty does not cover

- Damage from pumping sand, cement, saltwater, chemicals or liquids above 35 degrees C.
- Damage from running the pump dry for long periods, or from frost.
- Damage caused by opening the motor housing or by unauthorised repairs.
- Commercial rental use.

How to make a claim

Contact AquaFlow Service with your proof of purchase and the serial number. Do not send the pump without a return authorisation (RMA) number.

- Phone: +1 555 0199 2200 (Monday to Friday, 08:00 to 18:00)
- Email: service@aquaflow.example

Disposal

At the end of its life, do not dispose of the pump with household waste. Take it to an electrical waste collection point.